Statistical Analysis of Consumer Spending on Beauty Products

PROJECT 1

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NEGAR ORANGI-FARD | DATA ANALYSIS FOR SCIENCE – SECTION 07

Part 1: Preliminary Discussion and Data Collection

This study examines how much money the average woman spends on beauty supply products per month. The goal is to analyze consumer spending habits, identify trends, and determine the most common beauty product categories that women prioritize in their budgets. As a new small business owner, I'm interested in understanding whether the beauty supply industry is a profitable market to invest in. By analyzing how much the average women spends on cosmetic products, I can gain valuable insights into consumer spending habits and market demand. This research will allow me to identify price ranges that appeal to consumers and determine whether certain beauty categories such as makeup, skincare, haircare, and nails are more lucrative than others.

The specific population for this study is women who are between the ages of 18 and 38, who purchases beauty products in Atlanta, GA. The data collected includes:

- Amount spent on beauty products per month (Quantitative Variable #1)
- Number of beauty products purchased per month (Quantitative Variable #2)
- Type of beauty products most frequently purchased (Qualitative Variable)

By focusing on women aged 18-38 years old, the study ensures a representative sample of consumers who actively participate in the beauty market. This helps capture a range of spending behaviors across different age groups.

For this study, stratified sampling was chosen as the sampling method to ensure a balanced representation across different age groups. Stratified sampling is useful when a population has distinct subgroups that might have different spending habits. Since women of different ages may spend differently on beauty products, this method helps reduce bias results.

1. The population was divided into five age-based groups
    a. 18-21 years old
    b. 22-25 years old
    c. 26-29 years old
    d. 30-33 years old
    e. 34-38 years old
2. From each age group, a proportional number of participants were randomly selected to ensure fair representation.
3. Women from each group were asked how much they spend on beauty products, how many products they purchase per month and which category they spend the most on.

This approach allowed for a fair and balanced dataset ensuring that spending costs across different age groups were captured correctly.

Part 2: Record and Summarize Data

Table 1:
The dataset consists of 20 women who provided information on their beauty product spending habits. The table includes the following three variables:

Column 1: Monthly Spending – Quantitative
Column 2: Number of Products Purchased – Quantitative
Column 3: Category of Beauty Products Spent the Most – Qualitative

Amount Spent on Beauty Products per Month ($)	Number of Beauty Products Purchased per Month	Type of Beauty Products Spent on the Most
45	3	Skincare
75	5	Makeup
60	4	Haircare
100	6	Skincare
30	2	Haircare
120	7	Makeup
55	3	Nails
80	5	Skincare
90	6	Makeup
25	2	Haircare
70	4	Skincare
150	8	Makeup
40	3	Nails
110	6	Haircare
35	2	Skincare
95	5	Makeup
85	5	Haircare
20	1	Skincare
130	7	Makeup
50	3	Skincare

Table 2:
The dataset consists of 20 responses with spending amounts ranging from $0-$150 per month. Below are the results of the quantitative variables grouped together:

Amount Spent ($)	Frequency	Relative Frequency (%)
$0-$30	2	10.0
$31-$60	5	25.0
$61-$90	6	30.0
$91-$120	5	25.0
$121-$150	2	10.0

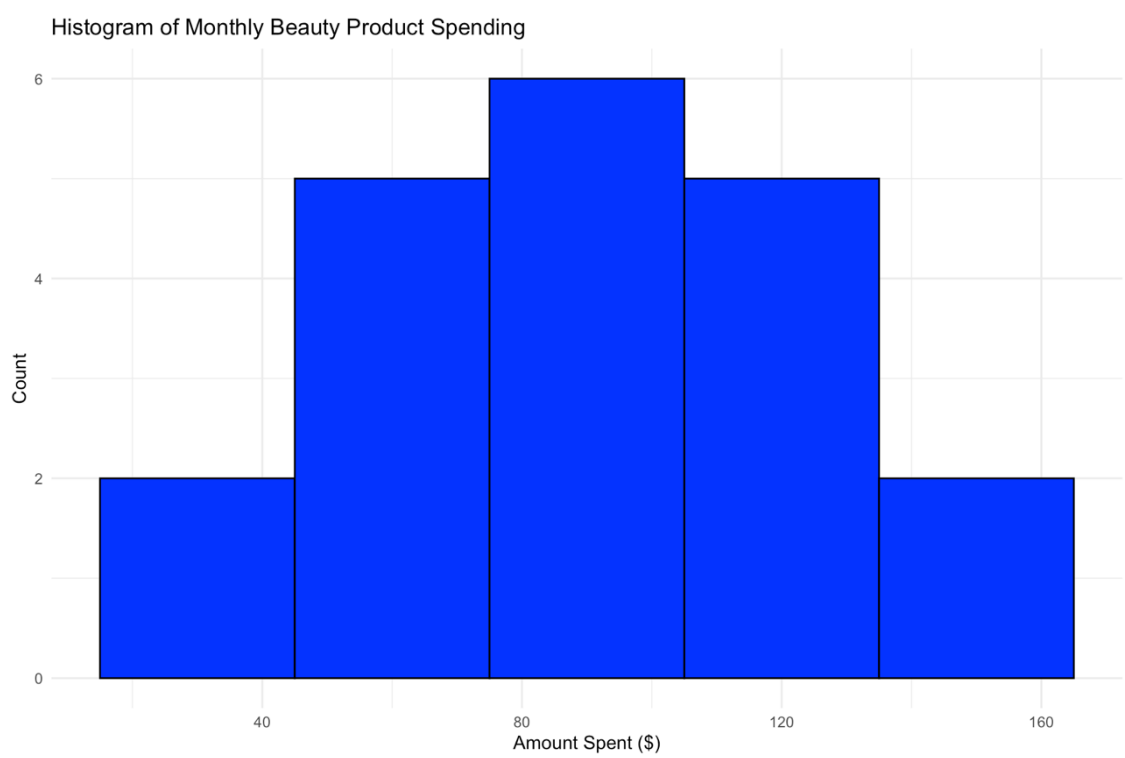
Table 3:
The dataset below consists of the qualitative variable that represents the type of beauty products women spend the most on including makeup, skincare, haircare, and nails:

Beauty Product	Frequency	Relative Frequency (%)
Skincare	6	30.0
Makeup	6	30.0
Haircare	5	25.0
Nails	3	15.0

Part 3: Descriptive Statistics

A histogram was created using RStudio to visually represent the distribution of the data on how much women spend on beauty supply products per month. The bars represent the frequency which makes it easier to see which spending ranges are most common:

Given dataset: 45, 75, 60, 100, 30, 120, 55, 80, 90, 25, 70, 150, 40, 110, 35, 95, 85, 20, 130, 50



Minimum Value:
To find the minimum value,
I rearranged the dataset from smallest to largest and identified the smallest number in the set. This was also calculated in RStudio.

20, 25, 30, 35, 40, 45, 50, 55, 60, 70, 75, 80, 85, 90, 95, 100, 110, 120, 130, 150

Minimum = $20.

Mean Value:
To find the mean of the dataset,
I added all 20 values together then divided by 20 to get the mean.

$$\text{mean} = \frac{\text{sum of all data values}}{\text{total number of values}}$$

mean = 45+75+60+100+30+120+55+80+90+25+70+150+40+110+35+95+85+20+130+50 = 1475

$$\text{mean} = \frac{1475}{20} = 73.75$$

Mean = $73.25 or $73.00

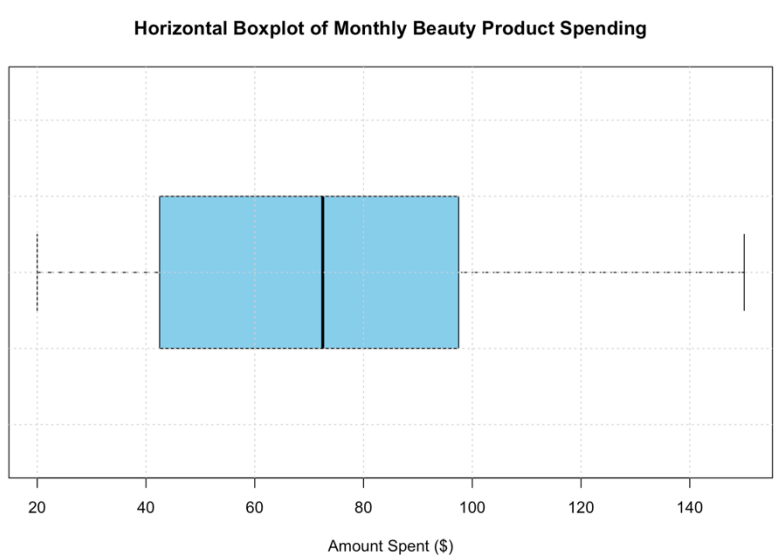
Mode Value:
The mode is the number that appears most frequently in the dataset.

20, 25, 30, 35, 40, 45, 50, 55, 60, 70, 75, 80, 85, 90, 95, 100, 110, 120, 130, 150

Since each number only appears once, there is no mode.

Mode = none

A boxplot was created in RStudio to visually summarize key statistics including the median, quartiles, and potential outliers:



Q1 (First Quartile) = $43.75
25% of women spent $43.75 or less per month on beauty products.

Q3 (Third Quartile) = $96.25
75% of women spent $96.25 or less per month.

Range = $130
To find the range, I subtracted the maximum value from the minimum value.

Range = Maximum − Minimum
Range = 150 – 20 = 130

Standard Deviation = $36.82
RStudio was used to calculate the standard deviation.

Part 4: Describe and Analyze Data

1. Discuss the symmetry of the histogram including comparisons of the mean and the median. What does this say about your data?
    - Based on the shape of the histogram and comparison between the mean ($73.25) and median ($72.50) the spending behavior is symmetrical.
2. What is the overall shape of the data set? Based on your answer, determine which measure of the center (mean, median, mode) is the best and why?
    - The data set is approximately symmetrical so the best measure of the center is the mean ($73.25) and median ($72.50) are very close. The mean would best measure of center since it represents the data best.
3. Does the middle 50% of the data appear to be concentrated together or spread apart? Explain how you determined this; also explain how you calculated the middle 50%.
    - The middle 50% of the data is measured using the Interquartile Range (IQR) which is calculated by Q3 - Q1, where Q1 = 25th percentile and Q3 is the 75th percentile. I used RStudio to calculate the IQR which is $52.50. This means the data appears to be more spread apart than closely together. I was able to use RStudio to determine this value.
4. Find 20th percentile (Show work). What do this value mean?
    - Using Rstudio, I was able to see that the 20th percentile is $39.00. This means that 20% of people in the data set spend less than $39.00 per month on beauty products, while 80% of people spend more than $39.00.
5. Find 95th percentile (Show work). What do this value mean?
    - Using RStudio, I was able to see that the 95th percentile is $131.00. This means that 95% of people in the data set spends less than $131.00 per month on beauty products, while only 5% spend more than $131.00.
6. Find Interquartile range. Show work. What does the Interquartile range IQR say about your data?

    - The Interquartile Range (IQR) is $52.50, which was calculated by Q3 ($43.75) – Q1 ($96.25).
7. Find the interval that contains the data within two standard deviations of your mean. Show work.

- The interval within two standard deviations of the mean is [0.40 and 146.90]. This was calculated using RStudio. This means that 95% of the data falls between $0.40 and $146.90.

8. Did you notice any potential outliers? If so, what values are they? Show your work in how you used the potential outlier formula to determine whether the values might be outliers.
    - There are no potential outliers in the data set because all values fall within the range [-35,175]. Using RStudio, the Interquartile Range (IQR) method was used to check for outliers.

Conclusion:

As a new small business owner, I'm excited about the opportunity to invest in the beauty supply industry. Based on the analysis this industry has great potential. On average consumers spend $73.25 per month on beauty products with the top 5% spending over $131 showing a high demand. The range of spending habits means there is a chance to attract both budget-friendly and high-spending customers. Since there are no extreme outliers the spending patterns seem pretty steady and predictable making it easier to plan for success. With the right pricing and marketing strategy this industry has huge profit potential and I'm excited to move forward with this investment.

Resources – Rstudio Syntax

Table 1:
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Column 1: Monthly Spending – Quantitative
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25	2	Haircare
70	4	Skincare
150	8	Makeup
40	3	Nails
110	6	Haircare
35	2	Skincare
95	5	Makeup
85	5	Haircare
20	1	Skincare
130	7	Makeup
50	3	Skincare

This table was created using Excel Spreadsheet.

Table 2: The dataset consists of 20 responses, with spending amounts ranging from $20-$150 per month. Below are the results of the quantitative variables grouped together:

Amount Spent ($)	Frequency	Relative Frequency (%)
$0-$30	2	10.0
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$121-$150	2	10.0

This table was created using Excel Spreadsheet.

Table 3: The dataset below consists of the qualitative variable that represents the type of beauty products women spend the most on including makeup, skincare, haircare, and nails:

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Makeup	6	30.0
Haircare	5	25.0
Nails	3	15.0

This table was created using Excel Spreadsheet.

A histogram was created using RStudio to visually represent the distribution of the data on how much women spend on beauty supply products per month. The bars represent the frequency which makes it easier to see which spending ranges are most common:

```
Console   Terminal   Background Jobs
R 4.4.2 · ~/Project 1 - Data Collection & Descriptive Statistics/

R version 4.4.2 (2024-10-31) -- "Pile of Leaves"
Copyright (C) 2024 The R Foundation for Statistical Computing
Platform: aarch64-apple-darwin20

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

  Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> install.packages("ggplot2")  # Run this only if ggplot2 is not installed
trying URL 'https://cran.rstudio.com/bin/macosx/big-sur-arm64/contrib/4.4/ggplot2_3.5.1.tgz'
Content type 'application/x-gzip' length 4974305 bytes (4.7 MB)
==================================================
downloaded 4.7 MB


The downloaded binary packages are in
        /var/folders/7h/hhx84nv124q8y01kd0c8xxr00000gn/T//Rtmp5RKM6g/downloaded_packages
> library(ggplot2)
>
> spending_data <- read.csv(file.choose(), stringsAsFactors = FALSE)  # Manually select file
>
> # Function to calculate midpoints from range values
> get_midpoint <- function(range) {
+     values <- as.numeric(unlist(strsplit(range, "-")))  # Split the range
+     return(mean(values, na.rm = TRUE))  # Compute the midpoint
+ }
>
> # Apply function to create a numeric "Midpoint" column
> spending_data$Spending_Midpoint <- sapply(spending_data$Amount.Spent, get_midpoint)
>
> # Convert "Frequency" column to numeric
> spending_data$Frequency <- as.numeric(spending_data$Frequency)
>
> ggplot(spending_data, aes(x = Spending_Midpoint, weight = Frequency)) +
+     geom_histogram(binwidth = 30, fill = "blue", color = "black") +
+     labs(title = "Histogram of Monthly Beauty Product Spending",
+          x = "Amount Spent ($)",
+          y = "Count") +
+     theme_minimal()
>
>
```

Minimum Value:
To find the minimum value,
I rearranged the dataset from smallest to largest and identified the smallest number in the set. This was also calculated in Rstudio.

20,25,30,35,40,45,50,55,60,70,75,80,85,90,95,100,110,120,130,150

Minimum = $20.

Mean Value:
To find the mean of the dataset,
I added all 20 values together then divided by 20 to get the mean.

$$\text{mean} = \frac{\text{sum of all data values}}{\text{total number of values}}$$

mean = 45+75+60+100+30+120+55+80+90+25+70+150+40+110+35+95+85+20+130+50 = 1475

$$\text{mean} = \frac{1475}{20} = 73.75$$

Mean = $73.25 or $73.00

Mode Value:
The mode is the number that appears most frequently in the dataset.
I tried to calculate mode in RStudio but it kept giving me different numbers so I just calculated it myself by find the most common number which out of this dataset there were no repeating numbers.

20,25,30,35,40,45,50,55,60,70,75,80,85,90,95,100,110,120,130,150

Since each number only appears once, there is no mode.

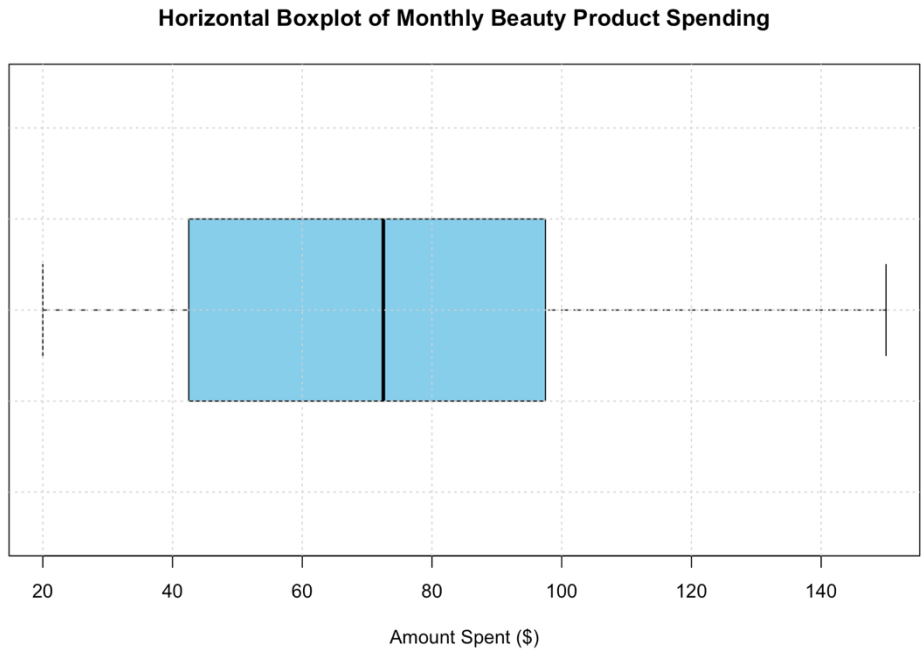
Mode = none

Console | Terminal | Background Jobs

R 4.4.2 · ~/Project 1 – Data Collection & Descriptive Statistics/

```
[1] 75.5
>
> get_mode <- function(x) {
+     uniq_values <- unique(x)
+     uniq_values[which.max(tabulate(match(x, uniq_values)))]
+ }
>
> mode_value <- get_mode(spending_data$Spending_Midpoint)
> print(mode_value)
[1] 15
>
> cat("Minimum:", min_value, "\n")
Minimum: 15
> cat("Mean:", mean_value, "\n")
Mean: 75.4
> cat("Median:", median_value, "\n")
Median: 75.5
> cat("Mode:", mode_value, "\n")
Mode: 15
>
> # Define the dataset as a numeric vector
> spending_data <- c(45, 75, 60, 100, 30, 120, 55, 80, 90, 25,
+                    70, 150, 40, 110, 35, 95, 85, 20, 130, 50)
>
> min_value <- min(spending_data)
> print(min_value)
[1] 20
>
> mean_value <- mean(spending_data)
> print(mean_value)
[1] 73.25
>
> median_value <- median(spending_data)
> print(median_value)
[1] 72.5
>
> get_mode <- function(x) {
+     uniq_values <- unique(x)
+     uniq_values[which.max(tabulate(match(x, uniq_values)))]
+ }
>
> mode_value <- get_mode(spending_data)
> print(mode_value)
[1] 45
>
> cat("Minimum:", min_value, "\n")
Minimum: 20
> cat("Mean:", mean_value, "\n")
Mean: 73.25
> cat("Median:", median_value, "\n")
Median: 72.5
> cat("Mode:", mode_value, "\n")
Mode: 45
>
>
```

A boxplot was created in Rstudio to visually summarize key statistics including the median, quartiles, and potential outliers:




```
>
> # Define the dataset as a numeric vector
> spending_data <- c(20, 25, 30, 35, 40, 45, 50, 55, 60, 70,
+                    75, 80, 85, 90, 95, 100, 110, 120, 130, 150)
>
> # Create a horizontal boxplot
> boxplot(spending_data,
+         main = "Horizontal Boxplot of Monthly Beauty Product Spending",
+         xlab = "Amount Spent ($)",
+         col = "skyblue", border = "black",
+         horizontal = TRUE)
>
> # Add grid lines for better readability
> grid()
>
>
```

9. Does the middle 50% of the data appear to be concentrated together or spread apart? Explain how you determined this; also explain how you calculated the middle 50%.
    - The middle 50% of the data is measured using the Interquartile Range (IQR) which is calculated by Q3 - Q1, where Q1 = 25th percentile and Q3 is the 75th percentile. I used RStudio to calculate the IQR which is $52.50. This means the data appears to be more spread apart than closely together.

```
>
> # Define dataset
> spending_data <- c(45, 75, 60, 100, 30, 120, 55, 80, 90, 25,
+                    70, 150, 40, 110, 35, 95, 85, 20, 130, 50)
>
> # Compute Q1 and Q3
> q1 <- quantile(spending_data, 0.25)  # First quartile (Q1)
> q3 <- quantile(spending_data, 0.75)  # Third quartile (Q3)
>
> # Compute IQR
> iqr_value <- q3 - q1
>
> # Display results
> cat("Q1:", q1, "\n")
Q1: 43.75
> cat("Q3:", q3, "\n")
Q3: 96.25
> cat("Interquartile Range (IQR):", iqr_value, "\n")
Interquartile Range (IQR): 52.5
>
>
```

10. Find 20th percentile (Show work). What do this value mean?
    - Using Rstudio, I was able to see that the 20th percentile is $39.00. This means that 20% of people in the dataset spend less than $39.00 per month on beauty products, while 80% of people spend more than $39.00.

```
>
> # Define the dataset
> spending_data <- c(45, 75, 60, 100, 30, 120, 55, 80, 90, 25,
+                    70, 150, 40, 110, 35, 95, 85, 20, 130, 50)
>
> # Compute the 20th percentile
> p20 <- quantile(spending_data, 0.20)
>
> # Display the result
> cat("20th Percentile (P20):", p20, "\n")
20th Percentile (P20): 39
>
>
```

11. Find 95th percentile (Show work). What do this value mean?
    - Using RStudio, I was able to see that the 95th percentile is $131.00. This means that 95% of people in the data set spends less than $131.00 per month on beauty products, while only 5% spend more than $131.00.

```
> # Define the dataset
> spending_data <- c(45, 75, 60, 100, 30, 120, 55, 80, 90, 25,
+                    70, 150, 40, 110, 35, 95, 85, 20, 130, 50)
>
> # Compute the 95th percentile
> p95 <- quantile(spending_data, 0.95)
>
> # Display the result
> cat("95th Percentile (P95):", p95, "\n")
95th Percentile (P95): 131
>
>
```

12. Find the interval that contains the data within two standard deviations of your mean. Show work.
    - The interval within two standard deviations of the mean is [0.40 and 146.90]. This was calculated using RStudio. This means that 95% of the data falls between $0.40 and $146.90.

```
> # Define the dataset
> spending_data <- c(45, 75, 60, 100, 30, 120, 55, 80, 90, 25,
+                    70, 150, 40, 110, 35, 95, 85, 20, 130, 50)
>
> # Compute Mean and Standard Deviation
> mean_value <- mean(spending_data)
> std_dev <- sd(spending_data)
>
> # Compute Interval Within Two Standard Deviations
> lower_bound <- mean_value - (2 * std_dev)
> upper_bound <- mean_value + (2 * std_dev)
>
> # Display Results
> cat("Mean:", mean_value, "\n")
Mean: 73.25
> cat("Standard Deviation:", std_dev, "\n")
Standard Deviation: 36.82373
> cat("Interval within Two Standard Deviations: [", lower_bound, ",", upper_bound, "]\n")
Interval within Two Standard Deviations: [ -0.3974532 , 146.8975 ]
>
>
```

13. Did you notice any potential outliers? If so, what values are they? Show your work in how you used the potential outlier formula to determine whether the values might be outliers.
    - There are no potential outliers in the data set because all values fall within the range [-35,175]. Using RStudio, the Interquartile Range (IQR) method was used to check for outliers.

Console | Terminal | Background Jobs

R 4.4.2 · ~/Project 1 – Data Collection & Descriptive Statistics/

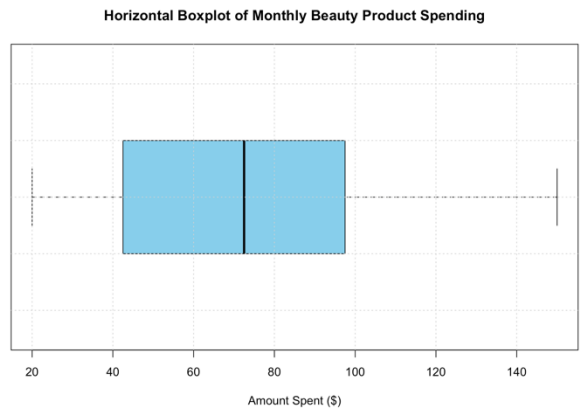
```
Q3: 96.25
> cat("Interquartile Range (IQR):", iqr_value, "\n")
Interquartile Range (IQR): 52.5
>
> # Define the dataset
> spending_data <- c(45, 75, 60, 100, 30, 120, 55, 80, 90, 25,
+                    70, 150, 40, 110, 35, 95, 85, 20, 130, 50)
>
> # Compute Mean and Standard Deviation
> mean_value <- mean(spending_data)
> std_dev <- sd(spending_data)
>
> # Compute Interval Within Two Standard Deviations
> lower_bound <- mean_value - (2 * std_dev)
> upper_bound <- mean_value + (2 * std_dev)
>
> # Display Results
> cat("Mean:", mean_value, "\n")
Mean: 73.25
> cat("Standard Deviation:", std_dev, "\n")
Standard Deviation: 36.82373
> cat("Interval within Two Standard Deviations: [", lower_bound, ",", upper_bound, "]\n")
Interval within Two Standard Deviations: [ -0.3974532 , 146.8975 ]
>
> # Define the dataset
> spending_data <- c(45, 75, 60, 100, 30, 120, 55, 80, 90, 25,
+                    70, 150, 40, 110, 35, 95, 85, 20, 130, 50)
>
> # Compute Q1, Q3, and IQR
> q1 <- quantile(spending_data, 0.25)
> q3 <- quantile(spending_data, 0.75)
> iqr_value <- q3 - q1
>
> # Compute Outlier Boundaries
> lower_bound <- q1 - (1.5 * iqr_value)
> upper_bound <- q3 + (1.5 * iqr_value)
>
> # Identify potential outliers
> outliers <- spending_data[spending_data < lower_bound | spending_data > upper_bound]
>
> # Display Results
> cat("Q1:", q1, "\n")
Q1: 43.75
> cat("Q3:", q3, "\n")
Q3: 96.25
> cat("IQR:", iqr_value, "\n")
IQR: 52.5
> cat("Lower Bound:", lower_bound, "\n")
Lower Bound: -35
> cat("Upper Bound:", upper_bound, "\n")
Upper Bound: 175
> cat("Potential Outliers:", outliers, "\n")
Potential Outliers:
>
>
```

Environment | History | Connections | Tutorial

Import Dataset · 52 MiB · List

R · Global Environment

median_value	72.5
min_value	20
mode_value	45
outliers	numeric (empty)
p20	Named num 39
p95	Named num 131
q1	Named num 43.8
q3	Named num 96.2
range_value	130
spending_data	num [1:20] 45 75 60 100 30 120 55 80 90 25 ...
std_dev	36.8237266189241
upper_bound	Named num 175
Functions	
get_midpoint	function (range)
get_mode	function (x)

Files | Plots | Packages | Help | Viewer | Presentation

Zoom · Export · Publish